

EXTENSION 2

Complex Numbers (Ext2), N2 Using Complex Numbers (Ext2)

Solving Equations with Complex Numbers

Teacher: Nardene Dodd

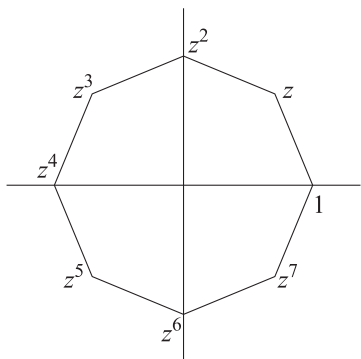
Exam Equivalent Time: 124.5 minutes (based on allocation of 1.5 minutes per mark)



Questions

1. Complex Numbers, EXT2 N2 2017 HSC 1 MC

The complex number z is chosen so that $1, z, \dots, z^7$ form the vertices of the regular polygon shown.



Which polynomial equation has all of these complex numbers as roots?

- A. $x^7 - 1 = 0$
- B. $x^7 + 1 = 0$
- C. $x^8 - 1 = 0$
- D. $x^8 + 1 = 0$

2. Complex Numbers, EXT2 N2 2018 HSC 6 MC

Which complex number is a 6th root of i ?

- A. $-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$
- B. $-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$
- C. $-\sqrt{2} + \sqrt{2}i$
- D. $-\sqrt{2} - \sqrt{2}i$

3. Complex Numbers, EXT2 N2 2016 HSC 10 MC

Suppose that $x + \frac{1}{x} = -1$.

What is the value of $x^{2016} + \frac{1}{x^{2016}}$?

- A. 1
- B. 2
- C. $\frac{2\pi}{3}$
- D. $\frac{4\pi}{3}$

4. Complex Numbers, EXT2 N2 2022 HSC 6 MC

It is known that a particular complex number z is NOT a real number.

Which of the following could be true for this number z ?

- A. $\bar{z} = iz$
- B. $\bar{z} = |z^2|$
- C. $\operatorname{Re}(iz) = \operatorname{Im}(z)$
- D. $\operatorname{Arg}(z^3) = \operatorname{Arg}(z)$

5. Complex Numbers, EXT2 N2 2016 HSC 12c

Let $z = \cos\theta + i\sin\theta$.

i. By considering the real part of z^4 , show that $\cos 4\theta$ is

$$\cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta. \quad (2 \text{ marks})$$

ii. Hence, or otherwise, find an expression for $\cos 4\theta$ involving only powers of $\cos\theta$. (1 mark)

6. Complex Numbers, EXT2 N2 2011 HSC 2c

Find, in modulus-argument form, all solutions of $z^3 = 8$. (2 marks)

7. Complex Numbers, EXT2 N2 2011 HSC 2d

i. Use the binomial theorem to expand $(\cos\theta + i\sin\theta)^3$. (1 mark)

ii. Use de Moivre's theorem and your result from part (i) to prove that

$$\cos^3\theta = \frac{1}{4}\cos 3\theta + \frac{3}{4}\cos\theta. \quad (3 \text{ marks})$$

iii. Hence, or otherwise, find the smallest positive solution of

$$4\cos^3\theta - 3\cos\theta = 1. \quad (2 \text{ marks})$$

8. Complex Numbers, EXT2 N2 EQ-Bank 1

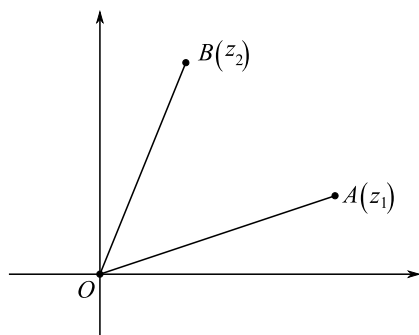
$z = \sqrt{2}e^{\frac{\pi}{10}}$ is a root of the equation $z^5 = \alpha(1 + i\sqrt{3})$, $\alpha \in \mathbb{R}$.

- Express $1 + i\sqrt{3}$ in exponential form. (2 marks)
- Find the value of α . (1 mark)
- Find the other 4 roots of the equation in exponential form. (3 marks)

9. Complex Numbers, EXT2 N2 2020 HSC 14a

Let z_1 be a complex number and let $z_2 = e^{\frac{4\pi}{3}} z_1$

The diagram shows points A and B which represent z_1 and z_2 , respectively, in the Argand plane.



- Explain why triangle OAB is an equilateral triangle. (2 marks)
- Prove that $z_1^2 + z_2^2 = z_1 z_2$. (3 marks)

10. Complex Numbers, EXT2 N2 2018 HSC 15b

- Use De Moivre's theorem and the expansion of $(\cos\theta + i\sin\theta)^8$ to show that

$$\begin{aligned} \sin 8\theta &= \binom{8}{1} \cos^7\theta \sin\theta - \binom{8}{3} \cos^5\theta \sin^3\theta \\ &\quad + \binom{8}{5} \cos^3\theta \sin^5\theta - \binom{8}{7} \cos\theta \sin^7\theta \end{aligned} \quad (2 \text{ marks})$$

- Hence, show that

$$\frac{\sin 8\theta}{\sin 2\theta} = 4(1 - 10\sin^2\theta + 24\sin^4\theta - 16\sin^6\theta). \quad (3 \text{ marks})$$

11. Complex Numbers, EXT2 N2 2006 HSC 2c

Find, in modulus-argument form, all solutions of $z^3 = -1$. (2 marks)

12. Complex Numbers, EXT2 N2 2009 HSC 2e

- Find all the 5th roots of -1 in modulus-argument form. (2 marks)
- Sketch the 5th roots of -1 on an Argand diagram. (1 mark)

13. Complex Numbers, EXT2 N2 2022 HSC 12e

Given the complex number $z = e^{i\theta}$, show that $w = \frac{z^2 - 1}{z^2 + 1}$ is purely imaginary. (3 marks)

14. Complex Numbers, EXT2 N2 2019 SPEC1-N 1

A cubic polynomial has the form $p(z) = z^3 + bz^2 + cz + d$, $z \in \mathbb{C}$, where $b, c, d \in \mathbb{R}$.

Given that a solution of $p(z) = 0$ is $z_1 = 3 - 2i$ and that $p(-2) = 0$, find the values of b, c and d . (4 marks)

15. Complex Numbers, EXT2 N2 2020 SPEC1 3

Find the cube roots of $\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$. Express your answers in modulus-argument form. (3 marks)

16. Complex Numbers, EXT2 N2 2023 HSC 12d

Find the cube roots of $2 - 2i$. Give your answer in exponential form. (3 marks)

17. Complex Numbers, EXT2 N2 SM-Bank 1

Find all solutions for z , in exponential form, given $z^4 = -2\sqrt{3} - 2i$. (3 marks)

18. Complex Numbers, EXT2 N2 SM-Bank 8

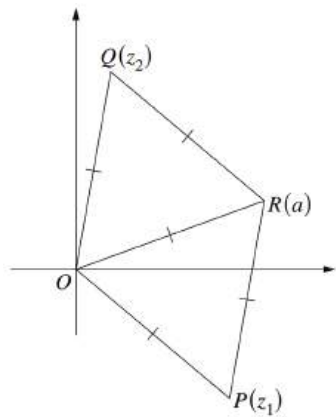
If $(x + iy)^3 = e^{-\frac{\pi}{2}}$, $x, y \in \mathbb{R}$, find a solution in the form $x + iy$, $x, y \neq 0$. (2 marks)

19. Complex Numbers, EXT2 N2 2021 HSC 14c

- Using de Moivre's theorem and the binomial expansion of $(\cos\theta + i\sin\theta)^5$, or otherwise, show that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$. (3 marks)

- By using part (i), or otherwise, show that $\operatorname{Re}\left(e^{\frac{i\pi}{10}}\right) = \sqrt{\frac{5 + \sqrt{5}}{8}}$. (3 marks)

20. Complex Numbers, EXT2 N2 2007 HSC 2d



The points P, Q and R on the Argand diagram represent the complex numbers z_1, z_2 and a respectively.

The triangles OPR and OQR are equilateral with unit sides, so $|z_1| = |z_2| = |a| = 1$.

Let $\omega = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$.

- Explain why $z_2 = \omega a$. (1 mark)
- Show that $z_1 z_2 = a^2$. (1 mark)
- Show that z_1 and z_2 are the roots of $z^2 - az + a^2 = 0$. (2 marks)

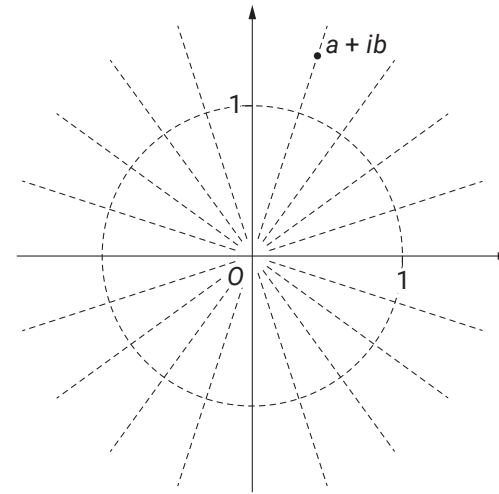
21. Complex Numbers, EXT2 N2 2022 HSC 13c

Consider the equation $z^5 + 1 = 0$, where z is a complex number.

- Solve the equation $z^5 + 1 = 0$ by finding the 5th roots of -1 . (2 marks)
- Show that if z is a solution of $z^5 + 1 = 0$ and $z \neq -1$, then $u = z + \frac{1}{z}$ is a solution of $u^2 - u - 1 = 0$. (2 marks)
- Hence find the exact value of $\cos \frac{3\pi}{5}$. (3 marks)

22. Complex Numbers, EXT2 N2 2021 HSC 13a

Indicate the locations of all of the fourth roots of the complex number $a + ib$.



23. Complex Numbers, EXT2 N2 2014 HSC 15b

- Using de Moivre's theorem, or otherwise, show that for every positive integer n ,

$$(1 + i)^n + (1 - i)^n = 2(\sqrt{2})^n \cos \frac{n\pi}{4}. \quad (2 \text{ marks})$$

- Hence, or otherwise, show that for every positive integer n divisible by 4,

$$\binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \dots + \binom{n}{n} = (-1)^{\frac{n}{4}} (\sqrt{2})^n \quad (3 \text{ marks})$$

24. Complex Numbers, EXT2 N2 2019 HSC 16b

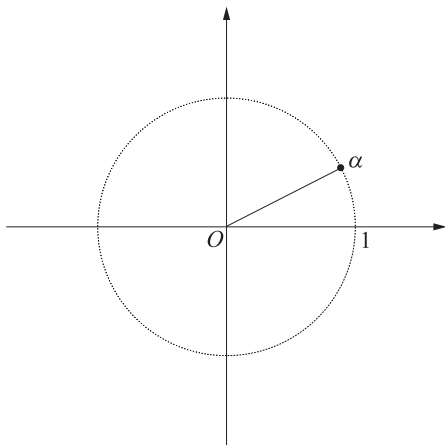
Let $P(z) = z^4 - 2kz^3 + 2k^2z^2 + mz + 1$, where k and m are real numbers.

The roots of $P(z)$ are $\alpha, \bar{\alpha}, \beta, \bar{\beta}$.

It is given that $|\alpha| = 1$ and $|\beta| = 1$.

i. Show that $(\operatorname{Re}(\alpha))^2 + (\operatorname{Re}(\beta))^2 = 1$. (3 marks)

ii. The diagram shows the position of α .



On the diagram, accurately show all possible positions of β . (2 marks)

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Worked Solutions

1. Complex Numbers, EXT2 N2 2017 HSC 1 MC

$P(x)$ has 8 separate roots.

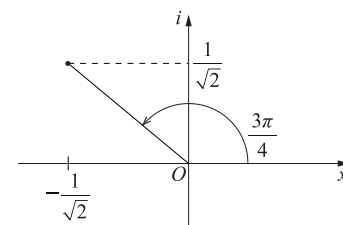
$\therefore$ Must be of degree at least 8.

Since 1 is also a root,

$\Rightarrow C$

2. Complex Numbers, EXT2 N2 2018 HSC 6 MC

Consider option A:



$$\left| -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right| = \sqrt{\left(-\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = 1$$

$$\arg(z) = \frac{3\pi}{4}$$

$$z = 1 \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

$$z^6 = \cos \frac{18\pi}{4} + i \sin \frac{18\pi}{4} \quad (\text{De Moivre})$$

$$= i$$

$\Rightarrow A$

3. Complex Numbers, EXT2 N2 2016 HSC 10 MC

$$x + \frac{1}{x} = -1$$

$$x^2 + x + 1 = 0$$

$$\therefore x = \frac{-1 \pm \sqrt{1-4}}{2}$$

$$= -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

$$|x| = \sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = 1$$

$$\therefore x = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$$

$$= \text{cis} \frac{2\pi}{3}$$

or

$$x = \cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3}$$

$$= \text{cis} \left(-\frac{2\pi}{3} \right)$$

$$x^{2016} = \text{cis}(\pm 1344\pi) \quad (\text{De Moivre})$$

$$= \text{cis} 0$$

$$= 1$$

$$\therefore x^{2016} + \frac{1}{x^{2016}} = 2$$

$$\Rightarrow B$$

♦ Mean mark 43%.

4. Complex Numbers, EXT2 N2 2022 HSC 6 MC

$$z \in \mathbb{C}, z \notin \mathbb{R}$$

$$\text{Let } z = a + ib \Rightarrow \bar{z} = a - ib$$

$$iz = i(a + ib) = -b + ia$$

$$\text{If } \bar{z} = iz,$$

$$a - ib = -b + ia$$

$$a = -b \quad (\text{satisfies})$$

$$\therefore \exists a, b (b \neq 0) \text{ such that } \bar{z} = iz$$

$$\Rightarrow A$$

♦♦ Mean mark 31%.

5. Complex Numbers, EXT2 N2 2016 HSC 12c

$$\text{i. } z = \cos \theta + i \sin \theta$$

$$z^4 = (\cos \theta + i \sin \theta)^4$$

$$= \cos^4 \theta + 4 \cos^3 \theta \cdot (i \sin \theta) + 6 \cos^2 \theta \cdot (i \sin \theta)^2 + 4 \cos \theta \cdot (i \sin \theta)^3 + (i \sin \theta)^4$$

$$= \cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4i \cos \theta \sin^3 \theta + \sin^4 \theta$$

$$z^4 = \cos 4\theta + i \sin 4\theta \quad (\text{by De Moivre})$$

Equating real parts:

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

... as required

$$\text{ii. } \cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2$$

$$= \cos^4 \theta - 6 \cos^2 \theta + 6 \cos^4 \theta + 1 - 2 \cos^2 \theta + \cos^4 \theta$$

$$= 8 \cos^4 \theta - 8 \cos^2 \theta + 1$$

6. Complex Numbers, EXT2 N2 2011 HSC 2c

$$z^3 = 8$$

$$z^3 = 8[\cos(2k\pi) + i\sin(2k\pi)], \quad k = 0, \pm 1, \pm 2$$

$$z = 2 \left[\cos\left(\frac{2k\pi}{3}\right) + i\sin\left(\frac{2k\pi}{3}\right) \right] \quad (\text{De Moivre})$$

When $k = 0$,

$$\begin{aligned} z &= 2(\cos 0 + i\sin 0) \\ &= 2\text{cis } 0 \end{aligned}$$

When $k = 1$,

$$\begin{aligned} z &= 2 \left[\cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right) \right] \\ &= 2\text{cis } \frac{2\pi}{3} \end{aligned}$$

When $k = -1$,

$$\begin{aligned} z &= 2 \left[\cos\left(\frac{-2\pi}{3}\right) + i\sin\left(\frac{-2\pi}{3}\right) \right] \\ &= 2\text{cis}\left(\frac{-2\pi}{3}\right) \end{aligned}$$

7. Complex Numbers, EXT2 N2 2011 HSC 2d

i. $(\cos\theta + i\sin\theta)^3$

$$\begin{aligned} &= \sum_{k=0}^3 {}^3C_k (\cos\theta)^{3-k} (i\sin\theta)^k \\ &= \cos^3\theta + 3\cos^2\theta(i\sin\theta) + 3\cos\theta(i\sin\theta)^2 + (i\sin\theta)^3 \\ &= \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta \end{aligned}$$

ii. Using De Moivre's Theorem

$$(\cos\theta + i\sin\theta)^3 = \cos 3\theta + i\sin 3\theta$$

Equate real parts

$$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$$

$$\cos 3\theta = \cos^3\theta - 3\cos\theta(1 - \cos^2\theta)$$

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$$

$$4\cos^3\theta = \cos 3\theta + 3\cos\theta$$

$$\therefore \cos^3\theta = \frac{1}{4}\cos 3\theta + \frac{3}{4}\cos\theta \quad \dots \text{as required}$$

iii. If $4\cos^3\theta - 3\cos\theta = 1$

$$\Rightarrow \cos 3\theta = 1 \quad (\text{from part (ii)})$$

$$3\theta = 2k\pi$$

$$\therefore \theta = \frac{2k\pi}{3}$$

$\therefore$ Smallest positive solution occurs when

$$\theta = \frac{2\pi}{3} \quad (\text{i.e. when } k = 1)$$

8. Complex Numbers, EXT2 N2 EQ-Bank 1

i. $\beta = 1 + i\sqrt{3}$

$$|\beta| = \sqrt{1 + (\sqrt{3})^2} = 2$$

$$\arg(\beta) = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$$

$$\beta = 2e^{i\frac{\pi}{3}}$$

ii. $z = \sqrt{2}e^{i\frac{\pi}{15}}$

$$z^5 = \left(\sqrt{2}e^{i\frac{\pi}{15}}\right)^5$$

$$= \left(\sqrt{2}\right)^5 e^{i\frac{\pi}{15} \times 5}$$

$$= 4\sqrt{2}e^{i\frac{\pi}{3}}$$

$$\therefore \alpha = 2\sqrt{2}$$

iii. $\arg(z^5) = \frac{\pi}{3} + 2k\pi, \quad k = 0, \pm 1, \pm 2, \dots$

$$\arg(z) = \frac{\pi}{15} + \frac{2k\pi}{5}$$

$$k = 1: \arg(z) = \frac{\pi}{15} + \frac{2\pi}{5} = \frac{11\pi}{15}$$

$$k = -1: \arg(z) = \frac{\pi}{15} - \frac{2\pi}{5} = -\frac{\pi}{3}$$

$$k = 2: \arg(z) = \frac{\pi}{15} + \frac{4\pi}{5} = \frac{13\pi}{15}$$

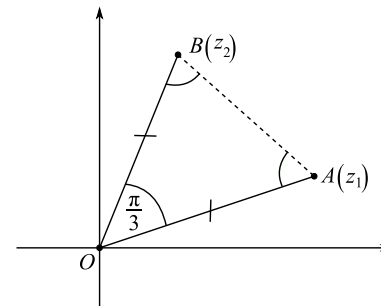
$$k = -2: \arg(z) = \frac{\pi}{15} - \frac{4\pi}{5} = -\frac{11\pi}{15}$$

$\therefore$ 4 other roots are:

$$e^{i\frac{11\pi}{15}}, e^{-i\frac{\pi}{3}}, e^{i\frac{13\pi}{15}}, e^{-i\frac{11\pi}{15}}$$

9. Complex Numbers, EXT2 N2 2020 HSC 14a

i.



Let $z_1 = r(\cos\theta + i\sin\theta)$

$$z_2 = e^{i\frac{\pi}{3}} z_1$$

$$= r\left(\cos\left(\theta + \frac{\pi}{3}\right) + i\sin\left(\theta + \frac{\pi}{3}\right)\right)$$

$$|z_1| = |z_2| \Rightarrow OA = OB$$

$$\angle AOB = \frac{\pi}{3} \quad \left(z_2 \text{ is a } \frac{\pi}{3} \text{ anti-clockwise rotation of } z_1\right)$$

$$\Rightarrow \angle OBA = \angle BAO = \frac{\pi}{3} \quad (\text{angles opposite equal sides})$$

$\therefore OAB$ is equilateral

ii. $z_1 = z_2 e^{i\frac{\pi}{3}}$

$$\frac{z_1}{z_2} = e^{i\frac{\pi}{3}}$$

$$\left(\frac{z_1}{z_2}\right)^3 = e^{(3 \times i\frac{\pi}{3})} \quad (\text{by De Moivre})$$

$$\frac{z_1^3}{z_2^3} = e^{i\pi}$$

$$z_1^3 = -z_2^3$$

$$z_1^3 + z_2^3 = 0$$

$$(z_1 + z_2)(z_1^2 - z_1 z_2 + z_2^2) = 0$$

$$z_1^2 - z_1 z_2 + z_2^2 = 0$$

$$z_1^2 + z_2^2 = z_1 z_2$$

COMMENT: The identity to prove suggests the addition of 2 cubes is a possible strategy.

10. Complex Numbers, EXT2 N2 2018 HSC 15b

i. By De Moivre

$$\cos\theta + i\sin\theta^8 = \cos 8\theta + i\sin 8\theta \dots (*)$$

Using Binomial Expansion

$$\begin{aligned} & (\cos\theta + i\sin\theta)^8 \\ &= \cos^8\theta + \binom{8}{1}\cos^7\theta \cdot i\sin\theta + \binom{8}{2}\cos^6\theta \cdot i^2\sin^2\theta \\ & \quad + \binom{8}{3}\cos^5\theta \cdot i^3\sin^3\theta + \binom{8}{4}\cos^4\theta \cdot i^4\sin^4\theta + \binom{8}{5}\cos^3\theta \cdot i^5\sin^5\theta \\ & \quad + \binom{8}{6}\cos^2\theta \cdot i^6\sin^6\theta + \binom{8}{7}\cos\theta \cdot i^7\sin^7\theta + i^8\sin^8\theta \end{aligned}$$

Equating imaginary parts of the expansion equation (*):

$$\begin{aligned} i\sin 8\theta &= \binom{8}{1}\cos^7\theta \cdot i\sin\theta + \binom{8}{3}i\cos^5\theta \cdot i^3\sin^3\theta \\ & \quad + \binom{8}{5}\cos^3\theta \cdot i^5\sin^5\theta + \binom{8}{7}\cos\theta \cdot i^7\sin^7\theta \\ \therefore \sin 8\theta &= \binom{8}{1}\cos^7\theta\sin\theta - \binom{8}{3}\cos^5\theta\sin^3\theta \\ & \quad + \binom{8}{5}\cos^3\theta\sin^5\theta - \binom{8}{7}\cos\theta\sin^7\theta \end{aligned}$$

$$\begin{aligned} \text{ii. } \sin 8\theta &= 8\cos^7\theta\sin\theta - 56\cos^5\theta\sin^3\theta + 56\cos^3\theta\sin^5\theta - 8\cos\theta\sin^7\theta \\ &= 2\sin\theta\cos\theta(4\cos^6\theta - 28\cos^4\theta\sin^2\theta + 28\cos^2\theta\sin^4\theta - 4\sin^6\theta) \end{aligned}$$

$$\begin{aligned} \therefore \frac{\sin 8\theta}{\sin 2\theta} &= 4\cos^6\theta - 28\cos^4\theta\sin^2\theta + 28\cos^2\theta\sin^4\theta - 4\sin^6\theta \\ &= 4(1 - \sin^2\theta)^3 - 28(1 - \sin^2\theta)^2\sin^2\theta + 28(1 - \sin^2\theta)\sin^4\theta - 4\sin^6\theta \\ &= 4(1 - 3\sin^2\theta + 3\sin^4\theta + \sin^6\theta) - 28\sin^2\theta(1 - 2\sin^2\theta + \sin^4\theta) \\ & \quad + 28\sin^4\theta(1 - \sin^2\theta) - 4\sin^6\theta \\ &= 4 - 40\sin^2\theta + 96\sin^4\theta - 56\sin^6\theta \\ &= 4(1 - 10\sin^2\theta + 24\sin^4\theta - 16\sin^6\theta) \end{aligned}$$

11. Complex Numbers, EXT2 N2 2006 HSC 2c

$$\begin{aligned} z^3 &= -1 \\ -1 &= \text{cis } (\pi + 2k\pi) \\ \therefore z &= \text{cis } \frac{\pi + 2k\pi}{3} \quad (\text{De Moivre}) \end{aligned}$$

When $k = 0$

$$z = \text{cis } \frac{\pi}{3}$$

When $k = 1$

$$z = \text{cis } \pi = -1$$

When $k = -1$

$$z = \text{cis } -\frac{\pi}{3}$$

$\therefore$ The 3 solutions to $z^3 = -1$ are

$$z = \text{cis } \frac{\pi}{3}, \quad -1, \quad \text{cis } -\frac{\pi}{3}$$

MARKER'S COMMENT: Another successful solution strategy was graphical, using the unit circle and three equal angles generated from pi.

12. Complex Numbers, EXT2 N2 2009 HSC 2e

$$\text{i. } z = \cos\theta + i\sin\theta$$

$$z^5 = \cos 5\theta + i\sin 5\theta = -1, \quad (\text{De Moivre})$$

$$\therefore \cos 5\theta = -1$$

$$5\theta = \pi, 3\pi, 5\pi, 7\pi, 9\pi$$

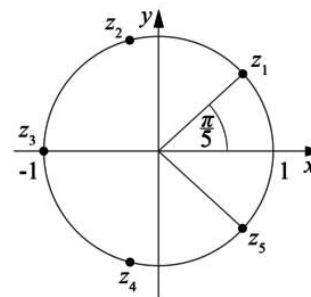
$$\theta = \frac{\pi}{5}, \frac{3\pi}{5}, \pi, \frac{7\pi}{5}, \frac{9\pi}{5}$$

$\therefore$ The roots are

$$z_1 = \text{cis } \frac{\pi}{5}, \quad z_2 = \text{cis } \frac{3\pi}{5}, \quad z_3 = \text{cis } \pi = -1,$$

$$z_4 = \text{cis } \frac{7\pi}{5}, \quad z_5 = \text{cis } \frac{9\pi}{5}$$

ii.



13. Complex Numbers, EXT2 N2 2022 HSC 12e

$$\begin{aligned}
z &= e^{i\theta} \\
w &= \frac{z^2 - 1}{z^2 + 1} \\
&= \frac{e^{i2\theta} - 1}{e^{i2\theta} + 1} \\
&= \frac{e^{i2\theta} - 1}{e^{i2\theta} + 1} \times \frac{e^{-i2\theta} + 1}{e^{-i2\theta} + 1} \\
&= \frac{1 + e^{i2\theta} - e^{-i2\theta} - 1}{1 + e^{i2\theta} + e^{-i2\theta} + 1} \\
&= \frac{e^{i2\theta} - e^{-i2\theta}}{2 + e^{i2\theta} + e^{-i2\theta}} \\
&= \frac{\cos(2\theta) + i\sin(2\theta) - (\cos(-2\theta) + i\sin(-2\theta))}{2 + \cos(2\theta) + i\sin(2\theta) + \cos(-2\theta) + i\sin(-2\theta)} \\
&= \frac{\cos(2\theta) + i\sin(2\theta) - \cos(2\theta) + i\sin(2\theta)}{2 + \cos(2\theta) + i\sin(2\theta) + \cos(2\theta) - i\sin(2\theta)} \\
&= \frac{i2\sin(2\theta)}{2 + 2\cos(2\theta)} \\
&= i \left(\frac{2\sin(2\theta)}{2 + 2\cos(2\theta)} \right) \quad (\text{purely imaginary})
\end{aligned}$$

14. Complex Numbers, EXT2 N2 2019 SPEC1-N 1

Roots: $z_1 = 3 - 2i$, $z_2 = 3 + 2i$, $z_3 = -2$

$$\begin{aligned}
p(z) &= (z - 3 + 2i)(z - 3 - 2i)(z + 2) \\
&= \left((z - 3)^2 - (2i)^2 \right) (z + 2) \\
&= (z^2 - 6z + 9 + 4)(z + 2) \\
&= (z^2 - 6z + 13)(z + 2) \\
&= z^3 + 2z^2 - 6z^2 - 12z + 13z + 26 \\
&= z^3 - 4z^2 + z + 26
\end{aligned}$$

$$\therefore b = -4, c = 1, d = 26$$

15. Complex Numbers, EXT2 N2 2020 SPEC1 3

$$z^3 = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i = \operatorname{cis}\left(-\frac{\pi}{4}\right) = \operatorname{cis}\left(-\frac{\pi}{4} + 2\pi k\right), k \in \mathbb{Z}$$

By De Moivre:

$$z = \operatorname{cis}\left(-\frac{\pi}{12} + \frac{2\pi k}{3}\right)$$

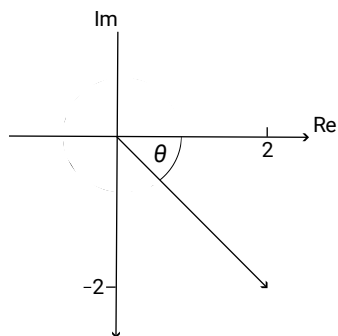
$$\text{If } k = 1, z_1 = \operatorname{cis}\left(-\frac{\pi}{12} + \frac{2\pi}{3}\right) = \operatorname{cis}\left(\frac{7\pi}{12}\right)$$

$$\text{If } k = 0, z_2 = \operatorname{cis}\left(-\frac{\pi}{12}\right)$$

$$\text{If } k = -1, z_3 = \operatorname{cis}\left(-\frac{\pi}{12} - \frac{2\pi}{3}\right) = \operatorname{cis}\left(\frac{-3\pi}{4}\right)$$

16. Complex Numbers, EXT2 N2 2023 HSC 12d

Express $2 - 2i$ in exponential form:



$$|2 - 2i| = 2\sqrt{2}$$

$$\arg(2 - 2i) = -\frac{\pi}{4}$$

$$2 - 2i = 2\sqrt{2}e^{-\frac{i\pi}{4}}$$

Find $z = re^{i\theta}$, where $z^3 = r^3e^{i3\theta} = 2\sqrt{2}e^{-\frac{i\pi}{4}}$

$$r^3 = 2\sqrt{2} \Rightarrow r = \sqrt{2}$$

$$i3\theta = -\frac{i\pi}{4}$$

$$\theta_1 = -\frac{\pi}{12}$$

Since roots are found symmetrically around Argand diagram:

$$\theta_2 = -\frac{\pi}{12} + \frac{2\pi}{3} = \frac{7\pi}{12}$$

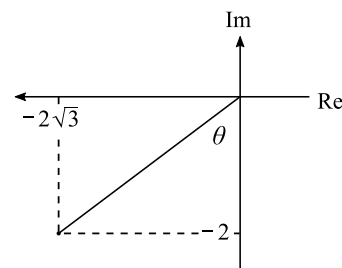
$$\theta_3 = -\frac{\pi}{12} - \frac{2\pi}{3} = -\frac{3\pi}{4}$$

$$\therefore \sqrt[3]{2 - 2i} = \sqrt{2}e^{-\frac{i\pi}{12}}, \sqrt{2}e^{\frac{7i\pi}{12}}, \sqrt{2}e^{-\frac{3i\pi}{4}}$$

17. Complex Numbers, EXT2 N2 SM-Bank 1

Convert z^4 to Mod/Arg:

$$|z^4| = \sqrt{(2\sqrt{3})^2 + 2^2} = 4$$



$$\tan \theta = \frac{2\sqrt{3}}{2} = \sqrt{3}$$

$$\theta = \frac{\pi}{3}$$

$$\arg(z^4) = -\left(\frac{\pi}{2} + \frac{\pi}{3}\right) = -\frac{5\pi}{6}$$

By De Moivre:

$$|z| = \sqrt[4]{4} = \sqrt{2}$$

$$\arg(z) = -\frac{5\pi}{24} + \frac{k\pi}{2}, \quad k = 0, \pm 1, \pm 2$$

$$k = 0: \arg(z) = -\frac{5\pi}{24}$$

$$k = 1: \arg(z) = -\frac{5\pi}{24} + \frac{\pi}{2} = \frac{7\pi}{24}$$

$$k = -1: \arg(z) = -\frac{5\pi}{24} - \frac{\pi}{2} = \frac{-17\pi}{24}$$

$$k = 2: \arg(z) = -\frac{5\pi}{24} + \pi = \frac{19\pi}{24}$$

$$\therefore \text{Solutions are: } \sqrt{2}e^{-\frac{5i\pi}{24}}, \sqrt{2}e^{\frac{-5i\pi}{24}}, \sqrt{2}e^{\frac{7i\pi}{24}}, \sqrt{2}e^{\frac{19i\pi}{24}}$$

18. Complex Numbers, EXT2 N2 SM-Bank 8

$$\text{Let } (x + iy) = \cos\theta + i\sin\theta$$

$$(x + iy)^3 = \cos(3\theta) + i\sin(3\theta)$$

$$e^{-\frac{4\pi}{3}} = \cos\left(\frac{-\pi}{2}\right) + i\sin\left(\frac{-\pi}{2}\right)$$

Equating real parts:

$$3\theta = \frac{-\pi}{2}$$

$$\theta = \frac{-\pi}{6}$$

$$\begin{aligned}\therefore x + iy &= \cos\left(\frac{-\pi}{6}\right) + i\sin\left(\frac{-\pi}{6}\right) \\ &= \frac{\sqrt{3}}{2} - i\frac{1}{2}\end{aligned}$$

COMMENT: Read the question fully and save time – only one solution is required here.

19. Complex Numbers, EXT2 N2 2021 HSC 14c

$$\text{i. } (\cos\theta + i\sin\theta)^5 = \cos 5\theta + i\sin 5\theta \quad (\text{by De Moivre})$$

Using binomial expansion:

$$\begin{aligned}(\cos\theta + i\sin\theta)^5 &= \cos^5\theta + 5\cos^4\theta \cdot i\sin\theta + 10\cos^3\theta \cdot i^2\sin^2\theta + 10\cos^2\theta \cdot i^3\sin^3\theta \\ &\quad + 5\cos\theta \cdot i^4\sin^4\theta + i^5\sin^5\theta \\ &= \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta + i \quad (\text{imaginary part})\end{aligned}$$

Equating real parts:

$$\begin{aligned}\cos 5\theta &= \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta \\ &= \cos^5\theta - 10\cos^3\theta(1 - \cos^2\theta) + 5\cos\theta(1 - \cos^2\theta)\sin^2\theta \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + (5\cos\theta - 5\cos^3\theta)(1 - \cos^2\theta) \\ &= 11\cos^5\theta - 10\cos^3\theta + 5\cos\theta - 5\cos^3\theta - 5\cos^3\theta + 5\cos^5\theta \\ &= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta\end{aligned}$$

$$\text{ii. } \operatorname{Re}\left(e^{\frac{4\pi}{10}}\right) = \cos \frac{\pi}{10}$$

Using part a:

♦♦ Mean mark 31%.

$$16\cos^5\left(\frac{\pi}{10}\right) - 20\cos^3\left(\frac{\pi}{10}\right) + 5\cos\left(\frac{\pi}{10}\right) = \cos\left(\frac{5\pi}{10}\right) = \cos\left(\frac{\pi}{2}\right) = 0$$

Solutions to $\cos(5\theta) = 0$

$$5\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$\theta = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{5\pi}{10}, \dots$$

$$\text{Let } \cos\left(\frac{\pi}{10}\right) = x$$

$$16x^5 - 20x^3 + 5x = 0$$

$$16x^4 - 20x^2 + 5 = 0$$

$$x^2 = \frac{20 \pm \sqrt{20^2 - 4 \cdot 16 \cdot 5}}{2 \times 16}$$

$$= \frac{20 \pm \sqrt{80}}{32}$$

$$= \frac{5 \pm \sqrt{5}}{8}$$

$$x = \pm \sqrt{\frac{5 \pm \sqrt{5}}{8}}$$

$$\cos \frac{\pi}{10} > 0 \Rightarrow x = \sqrt{\frac{5 + \sqrt{5}}{8}}$$

$$1 > \cos \frac{\pi}{10} > \cos \frac{3\pi}{10} > 0$$

$$\therefore \operatorname{Re} \left(e^{\frac{\pi}{10}} \right) = \cos \frac{\pi}{10} = \sqrt{\frac{5 + \sqrt{5}}{8}}$$

20. Complex Numbers, EXT2 N2 2007 HSC 2d

i. Since $\triangle ORQ$ is equilateral, each angle is $\frac{\pi}{3}$ radians.

From $R(a)$:

$Q(z_2)$ is an anticlockwise rotation through $\frac{\pi}{3}$.

$$\therefore z_2 = a \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = \omega a.$$

ii. Solution 1

Similarly, $a = z_1 \omega$

$$z_1 = \frac{a}{\omega}$$

$$\begin{aligned} \therefore z_1 z_2 &= \frac{a}{\omega} \times \omega a \\ &= a^2 \end{aligned}$$

Solution 2

$P(z_1)$ is a clockwise rotation of $R(a)$ through $\frac{\pi}{3}$.

$$\therefore z_1 = \bar{\omega} a.$$

$$\begin{aligned} \therefore z_1 z_2 &= \bar{\omega} a \times \omega a \\ &= a^2 \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right) \times \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \\ &= a^2 \left(\cos^2 \frac{\pi}{3} + \sin^2 \frac{\pi}{3} \right) \\ &= a^2 \end{aligned}$$

iii. $z^2 - az + a^2 = 0$

Let the roots be α and β .

$$\alpha + \beta = -\frac{b}{a} = a$$

$$\alpha\beta = \frac{c}{a} = a^2$$

$$z_1 z_2 = a^2 \quad (\text{part (ii)})$$

$$\begin{aligned} z_1 + z_2 &= \bar{\omega} a + \omega a \\ &= \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} + \cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right) a \\ &= 2 \cos \frac{\pi}{3} \times a \\ &= 2 \times \frac{1}{2} \times a \\ &= a \end{aligned}$$

$\therefore z_1$ and z_2 are the roots of $z^2 - az + a^2 = 0$.

21. Complex Numbers, EXT2 N2 2022 HSC 13c

i. $z^5 + 1 = 0 \Rightarrow z^5 = -1$

$$z = e^{i\left(\frac{2k+1}{5}\right)}, \quad k \in \{0, 1, -1, 2, -2\}$$

$$\therefore z = e^{i\frac{\pi}{5}}, e^{i\frac{3\pi}{5}}, e^{-i\frac{\pi}{5}}, -1, e^{-i\frac{3\pi}{5}}$$

ii. $z^5 + 1 = (z + 1)(z^4 - z^3 + z^2 - z + 1)$

Given $z \neq 1$,

$$z^4 - z^3 + z^2 - z + 1 = 0$$

Divide by z^2 ($z \neq 0$)

$$z^2 - z + 1 - \frac{1}{z} + \frac{1}{z^2} = 0$$

$$z^2 + \frac{1}{z^2} - \left(z + \frac{1}{z}\right) + 1 = 0$$

$$z^2 + 2 + \frac{1}{z^2} - \left(z + \frac{1}{z}\right) - 1 = 0$$

$$\left(z + \frac{1}{z}\right)^2 - \left(z - \frac{1}{z}\right) - 1 = 0$$

Let $u = z + \frac{1}{z}$:

$$\therefore u^2 - u - 1 = 0$$

iii. $u^2 - u - 1 = 0$

By quadratic formula:

$$u = \frac{1 \pm \sqrt{1 - 4 \times 1 \times (-1)}}{2}$$

$$= \frac{1 \pm \sqrt{5}}{2}$$

$$z + \frac{1}{z} = \frac{1 \pm \sqrt{5}}{2}$$

$$e^{i\frac{3\pi}{5}} + e^{-i\frac{3\pi}{5}} = \frac{1 - \sqrt{5}}{2}, \quad \left(\cos \frac{3\pi}{5} < 0\right)$$

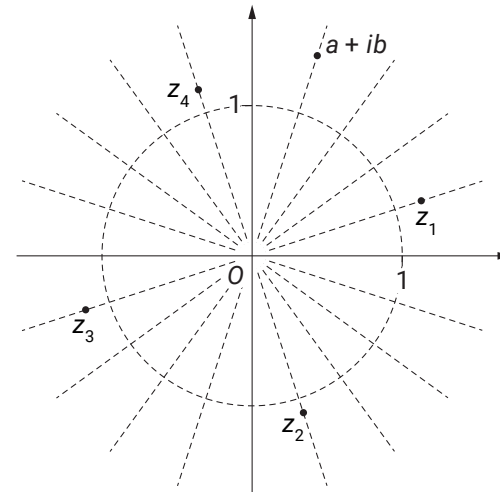
$$2\cos\left(\frac{3\pi}{5}\right) = \frac{1 - \sqrt{5}}{2}$$

$$\cos\left(\frac{3\pi}{5}\right) = \frac{1 - \sqrt{5}}{4}$$

Mean mark (ii) 53%.

♦ Mean mark (iii) 43%.

22. Complex Numbers, EXT2 N2 2021 HSC 13a



4 roots: z_1, z_2, z_3, z_4

$$\arg(z_1) = \frac{1}{4}\arg(a + ib)$$

$$|z| > 1 \text{ but less than } |a + ib|$$

$$\text{Rotations between roots} = \frac{\pi}{2}$$

♦ Mean mark 47%.

$$\begin{aligned}
 \text{i. } 1+i &= \sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right) \\
 &= \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \\
 (1+i)^n &= \left(\sqrt{2} \right)^n \left(\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right)
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 1-i &= \sqrt{2} \left(\cos \left(-\frac{\pi}{4} \right) + i \sin \left(-\frac{\pi}{4} \right) \right) \\
 &= \sqrt{2} \left(\cos \frac{\pi}{4} - i \sin \frac{\pi}{4} \right) \\
 (1-i)^n &= \left(\sqrt{2} \right)^n \left(\cos \frac{n\pi}{4} - i \sin \frac{n\pi}{4} \right) \\
 \therefore (1+i)^n + (1-i)^n &= \left(\sqrt{2} \right)^n \left(\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right) + \left(\sqrt{2} \right)^n \left(\cos \frac{n\pi}{4} - i \sin \frac{n\pi}{4} \right) \\
 &= \left(\sqrt{2} \right)^n \left(\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} + \cos \frac{n\pi}{4} - i \sin \frac{n\pi}{4} \right) \\
 &= 2 \left(\sqrt{2} \right)^n \cos \frac{n\pi}{4} \quad \dots \text{as required}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii. } (1+i)^n &= \binom{n}{0} + i \binom{n}{1} - \binom{n}{2} - i \binom{n}{3} + \binom{n}{4} + \dots + i^n \binom{n}{n} \\
 (1-i)^n &= \binom{n}{0} - i \binom{n}{1} - \binom{n}{2} + i \binom{n}{3} + \dots + (-1)^n i^n \binom{n}{n} \\
 (1+i)^n + (1-i)^n &= 2 \binom{n}{0} - 2 \binom{n}{2} + 2 \binom{n}{4} + \dots + 2 \binom{n}{n}
 \end{aligned}$$

Given n is divisible by 4, the last term $= + \binom{n}{n}$

♦♦ Mean mark 22%.
MARKER'S COMMENT: Few students could link part (i) to solving part (ii).

Using part (i):

$$\begin{aligned}
 2 \binom{n}{0} - 2 \binom{n}{2} + 2 \binom{n}{4} + \dots + 2 \binom{n}{n} &= 2 \left(\sqrt{2} \right)^n \cos \frac{n\pi}{4} \\
 \binom{n}{0} - \binom{n}{2} + \binom{n}{4} + \dots + \binom{n}{n} &= \left(\sqrt{2} \right)^n \cos \frac{n\pi}{4}
 \end{aligned}$$

If n is a positive integer divisible by 4 then $\cos \frac{n\pi}{4} = \pm 1$,
 depending on whether n is an even or odd multiple of 4.

$$\therefore \binom{n}{0} - \binom{n}{2} + \binom{n}{4} + \dots + \binom{n}{n} = (-1)^{\frac{n}{4}} \left(\sqrt{2} \right)^n$$

i. $P(z) = z^4 - 2kz^3 + 2k^2z^2 + mz + 1, \quad k, m \in \mathbb{R}$

Roots: $\alpha, \bar{\alpha}, \beta, \bar{\beta}$ and $|\alpha| = 1, |\beta| = 1$

Show $(\operatorname{Re}(\alpha))^2 + (\operatorname{Re}(\beta))^2 = 1$

$\alpha + \bar{\alpha} + \beta + \bar{\beta} = 2k$

$2\operatorname{Re}(\alpha) + 2\operatorname{Re}(\beta) = 2k$

$\operatorname{Re}(\alpha) + \operatorname{Re}(\beta) = k$

$\alpha\bar{\alpha} + \alpha\beta + \alpha\bar{\beta} + \bar{\alpha}\beta + \bar{\alpha}\bar{\beta} + \beta\bar{\beta} = 2k^2$

$|\alpha|^2 + \alpha(\beta + \bar{\beta}) + \bar{\alpha}(\beta + \bar{\beta}) + |\beta|^2 = 2k^2$

$1 + (\alpha + \bar{\alpha})(\beta + \bar{\beta}) + 1 = 2k^2$

$2 + 2\operatorname{Re}(\alpha) \cdot 2\operatorname{Re}(\beta) = 2(\operatorname{Re}(\alpha) + \operatorname{Re}(\beta))^2$

$2 + 4\operatorname{Re}(\alpha)\operatorname{Re}(\beta) = 2\operatorname{Re}(\alpha)^2 + 4\operatorname{Re}(\alpha)\operatorname{Re}(\beta) + 2\operatorname{Re}(\beta)^2$

$2 = 2(\operatorname{Re}(\alpha)^2 + \operatorname{Re}(\beta)^2)$

$\therefore 1 = \operatorname{Re}(\alpha)^2 + \operatorname{Re}(\beta)^2$

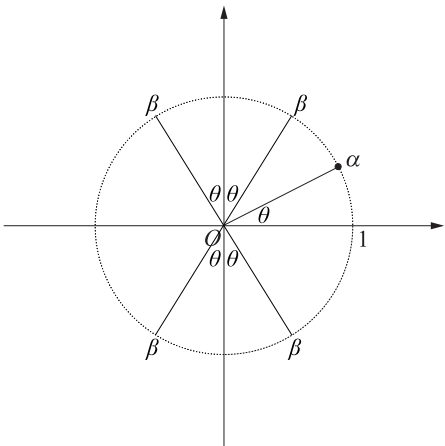
ii. $|\alpha| = |\beta|$ (given)

$\operatorname{Re}(\alpha)^2 + \operatorname{Re}(\beta)^2 = 1$ (see part (i))

$\operatorname{Re}(\alpha)^2 + \operatorname{Im}(\alpha)^2 = 1$ ($|\alpha| = 1$)

$\Rightarrow \operatorname{Re}(\beta)^2 = \operatorname{Im}(\alpha)^2$

$\operatorname{Re}(\beta) = \pm \operatorname{Im}(\alpha)$



♦♦ Mean mark part (i) 26%.

♦♦♦ Mean mark part (ii) 10%.